Huawei Access Node Hardware/Environmental Fault Resolution SOP (IND-ACC-58)

Node: IND-ACC-58 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_2629

Description

This Standard Operating Procedure outlines the steps to diagnose and resolve hardware or environmental faults causing ASIC memory parity errors, line card crashes, or loss of power supply redundancy on Huawei Access equipment at the Indiranagar Exchange site. The Huawei Access equipment is a critical component of the telecommunications network, playing a vital role in providing connectivity services to a large number of subscribers. The equipment's failure can lead to service outages, affecting the quality of service and customer experience. This SOP is designed to ensure timely and effective resolution of such faults, minimizing the impact on the network and services. The procedure involves a systematic approach to identifying the root cause of the fault, using a combination of diagnostic commands and physical checks. It outlines the steps to be taken to resolve the fault, including replacement of faulty components, reconfiguration of the equipment, and verification of the resolution. The SOP also includes rollback procedures to ensure that the network is restored to its original state in case the resolution fails. By following this SOP, the technical team can ensure that the Huawei Access equipment at the Indiranagar Exchange site operates within its designed specifications and provides reliable connectivity services to subscribers.

Pre-requisite Checks

- Dispatch remote hands to the Indiranagar Exchange site to ensure physical access to the equipment.

- Verify the redundant link status to ensure that it is active and functioning correctly.

- Check the environmental conditions at the site, including temperature, humidity, and power supply levels, to ensure that they are within the recommended operating range.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to hardware failure	High
Line card crashes due to software issues or hardware failure	Medium
Loss of power supply redundancy due to equipment failure or environmental conditions	Low

Diagnostic Commands & Expected Outputs

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Command: display device, Expected Output: Device information, including hardware version, software version, and configuration, Description: This command displays the device information, including hardware and software versions, configuration, and status. It helps to identify any hardware or software issues that may be causing the fault.
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Command: display board, Expected Output: Board information, including status, temperature, and power supply levels, Description: This command displays the board information, including status, temperature, and power supply levels. It helps to identify any issues with the line cards or power supplies that may be causing the fault.

Command: display power supply, Expected Output: Power supply information, including status, voltage, and current levels, Description: This command displays the power supply information, including status, voltage, and current levels. It helps to identify any issues with the power supplies that may be causing the fault.
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Resolution Steps

Step 1: Step: Replace the faulty ASIC or line card, Command: replace device asic [asic number], Description: This command replaces the faulty ASIC or line card with a new one. It ensures that the equipment operates within its designed specifications and provides reliable connectivity services to subscribers.

Step 2: Step: Reconfigure the equipment, Command: system reconfigure, Description: This command reconfigures the equipment, ensuring that it operates within its designed specifications and provides reliable connectivity services to subscribers.

Step 3: Step: Verify the resolution, Command: display device, Description: This command verifies the resolution by displaying the device information, including hardware and software versions, configuration, and status.

Post-Check Verification

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Command: display device, Expected Output: Device information, including hardware version, software version, and configuration, Description: This command verifies the resolution by displaying the device information, including hardware and software versions, configuration, and status.

Command: display board, Expected Output: Board information, including status, temperature, and power supply levels, Description: This command verifies the resolution by displaying the board information, including status, temperature, and power supply levels.

Command: display power supply, Expected Output: Power supply information, including status, voltage, and current levels, Description: This command verifies the resolution by displaying the power supply information, including status, voltage, and current levels.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: system restore, Description: This command restores the original configuration, ensuring that the network is restored to its original state in case the resolution fails.

- Step: Replace the faulty component with the original one, Command: replace device asic [asic number], Description: This command replaces the faulty component with the original one, ensuring that the network is restored to its original state in case the resolution fails.

- Step: Verify the rollback, Command: display device, Description: This command verifies the rollback by displaying the device information, including hardware and software versions, configuration, and status.